Sending Activation Tokens

The next step is to hook this up to our `registerUserHandler`, so that we generate an activation token when a user signs up and include it in their welcome email — similar to this:

```
Hi,

Thanks for signing up for a Greenlight account. We're excited to have you on board!

For future reference, your user ID number is 123.

Please send a request to the `PUT /v1/users/activated` endpoint with the following JSON
body to activate your account:

{"token": "Y3QMGX3PJ3WLRL2YRTQGQ6KRHU"}

Please note that this is a one-time use token and it will expire in 3 days.

Thanks,

The Greenlight Team
```

The most important thing about this email is that we're instructing the user to activate by issuing a `PUT` request to our API — not by *clicking a link* which contains the token as part of the URL path or query string.

Having a user click a link to activate via a `GET` request (which is used by default when clicking a link) would certainly be more convenient, but in the case of our API it has some big drawbacks. In particular:

- It would violate the HTTP principle that the `GET` method should only be used for 'safe' requests which retrieve resources — not for requests that modify something (like a user's activation status).
- It's possible that the user's web browser or antivirus will pre-fetch the link URL in the background, inadvertently activating the account. This Stack Overflow comment explains the risk of this nicely:

  > *That could result in a scenario where a malicious actor (Eve) wants to make an account using someone else's email (Alice). Eve signs up, and Alice received an email. Alice opens the email because she is curious about an account she didn't request. Her browser (or antivirus) requests the URL in the background, inadvertently activating the account.*

All-in-all, you should make sure that any actions which change the state of your application

(including activating a user) are only ever executed via `POST`, `PUT`, `PATCH` or `DELETE` requests — not by `GET` requests.

> **Note:** If your API is the back-end for a website, then you could adapt this email so that it asks the user to click a link which takes them to a page on your website. They can *then* click a button on the page to 'confirm their activation', which performs the `PUT` request to your API that actually activates the user. We'll look at this pattern in more detail in the next chapter.

But for now, if you're following along, go ahead and update your welcome email templates to include the activation token as follows: